

Possible VIVA Questions based on the ML Model

1. General Questions

Q1: What is the purpose of the FITSYNC ML model?

A: The FITSYNC ML model is designed for real-time heart rate anomaly detection. It uses an LSTM Autoencoder to learn normal heart rate patterns and identify irregularities based on reconstruction errors.

Q2: What type of data is used in this model?

A: The data consists of heart rate readings for different users, formatted as lists of time-series values. Each row in the dataset represents a user ID along with their recorded heart rates.

Q3: Why did you choose an LSTM Autoencoder for anomaly detection?

A:

- **LSTMs** excel at capturing temporal dependencies in sequential data like heart rates.
 - **Autoencoders** learn a compressed representation of normal patterns and reconstruct them. If a reconstructed sequence has high error, it indicates an anomaly.
 - This approach avoids manual threshold setting and adapts dynamically to individual users' heart rate trends.
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2. Data Preprocessing & Handling

Q4: Why did you apply normalization to the heart rate data?

A:

- Heart rates vary among individuals. Normalization scales all values between 0 and 1, ensuring uniform input distribution.
- It helps the model learn patterns independent of absolute values and prevents large values from dominating training.

Q5: How did you handle users with different amounts of heart rate data?

A:

- The data is grouped by `user_id`, and each user's heart rate values are normalized independently.
 - Sequences are created with a fixed length of 50 time steps to maintain consistency across users.
 - If sequences are too short, they are excluded to ensure proper learning.
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3. Model Architecture

Q6: Explain the architecture of the LSTM Autoencoder.

A:

- **Encoder:** Two-layer LSTM compresses input sequences into a latent vector.
- **Latent Space:** A fully connected (`Linear`) layer reduces dimensionality.
- **Decoder:** Two-layer LSTM reconstructs the original sequence from the latent representation.
- **Output Layer:** A `Linear` layer converts the decoder's output back to heart rate values.

Q7: Why do we use a bottleneck layer in the autoencoder?

A:

- The bottleneck layer forces the model to learn compressed patterns of normal heart rate behavior.
- Anomalous inputs, which do not fit these learned patterns, will have higher reconstruction errors.

4. Training & Optimization

Q8: What loss function is used and why?

A:

- **Mean Squared Error (MSE)** is used because it measures the difference between the original and reconstructed sequences.
- Higher reconstruction error indicates anomalies.

Q9: What optimization technique did you use and why?

A:

- **Adam Optimizer** is used due to its adaptive learning rate and efficiency in handling non-stationary data.
- It converges faster compared to traditional SGD.

Q10: How did you decide on the number of epochs and batch size?

A:

- **Epochs:** 5 epochs were chosen based on initial testing to balance training time and performance.
- **Batch Size:** 256 ensures efficient training while keeping memory usage manageable.

5. Model Deployment & Performance

Q11: Why did you export the model to ONNX?

A:

- ONNX allows cross-platform deployment, making the model compatible with MATLAB and edge devices.
- It optimizes inference speed and supports hardware acceleration.

Q12: How does the model detect anomalies in real-time?

A:

- A new heart rate sequence is fed into the trained model.
 - The reconstruction error is calculated. If it exceeds a threshold, it is flagged as an anomaly.
 - This enables real-time monitoring.
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6. Comparison with Other Models

Q13: Why not use a traditional statistical method like Z-score or IQR for anomaly detection?

A:

- These methods assume a normal distribution, which may not apply to heart rate data.
- They cannot capture time dependencies or gradual changes in patterns.

Q14: Why not use a simple RNN instead of LSTM?

A:

- RNNs suffer from the **vanishing gradient problem**, making them ineffective for long sequences.
- LSTMs use **gates** to retain important information across longer time steps.

Q15: Why not use a CNN-based model?

A:

- CNNs are good for spatial patterns but do not handle time dependencies well.
- LSTMs excel at learning from sequential data.

Q16: Why is this model better than supervised anomaly detection approaches?

A:

- Supervised models require labeled anomaly data, which is **hard to obtain for real-world heart rate data**.

- The LSTM Autoencoder is **unsupervised**, meaning it only learns from normal patterns and flags anything unusual.
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7. Future Improvements

Q17: How can this model be improved further?

A:

- **Threshold Tuning:** Dynamically adjust the anomaly threshold based on user history.
- **Hybrid Models:** Combine LSTM Autoencoders with Transformers for better anomaly detection.
- **Edge Deployment:** Optimize for real-time processing on smartwatches or IoT devices.

Q18: Can this model be used for other physiological signals?

A:

- Yes! It can be adapted for ECG, blood pressure, and respiratory rate by changing the input features.
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Why is this the Best Model?

✓ **Handles Real-Time Data Efficiently:** Works on live heart rate streams without requiring labeled anomaly data.

✓ **Captures Time Dependencies:** LSTMs outperform traditional methods in sequential pattern recognition.

✓ **Generalizable:** Works for multiple users without needing user-specific retraining.

✓ **Robust to Noise:** Autoencoder structure ensures small fluctuations do not trigger false positives.

✓ **Cross-Platform Deployment:** ONNX compatibility allows integration into MATLAB and IoT devices.

VIVA QUESTIONS FOR THE FITSYNC ML MODEL

1. General Questions

- **What is the primary objective of this model?**
→ Detecting heart rate anomalies in real-time using an LSTM Autoencoder.
 - **What dataset was used for training?**
→ A CSV file containing user IDs and corresponding heart rate sequences.
 - **Why was deep learning preferred over traditional statistical methods?**
→ Deep learning captures long-term temporal dependencies, making it better at detecting subtle anomalies.
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2. Preprocessing & Data Handling

- **Why is normalization important in this model?**
→ It ensures that all heart rates are scaled consistently, preventing large values from dominating training.
 - **What techniques were used for data normalization?**
→ Min-Max Scaling (scaling between 0 and 1 for each user's heart rate).
 - **Why did we apply sequence segmentation?**
→ LSTM models need fixed-length input sequences (50 time steps in this case).
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3. Model Architecture & Training

- **Why was an LSTM Autoencoder chosen?**
→ LSTM captures temporal dependencies, and autoencoders learn normal patterns, making them ideal for anomaly detection.
- **Explain the architecture of the model.**
→ Encoder (2-layer LSTM) → Latent space (Dense layer) → Decoder (2-layer LSTM) → Output (Reconstructed heart rate).
- **Why is Mean Squared Error (MSE) used as a loss function?**

→ MSE measures reconstruction error, helping identify anomalies when errors are large.

- **Why do we train on normal heart rate data only?**

→ The model learns normal patterns, so deviations in reconstruction signal anomalies.

4. Threshold Selection & Anomaly Detection

- **How is the anomaly threshold determined?**

→ Using a statistical percentile (e.g., 95th percentile of reconstruction error).

- **Why not use a fixed threshold for all users?**

→ Each user's normal heart rate range varies, requiring adaptive thresholds.

- **What happens if the threshold is too high or too low?**

→ Too high: Some anomalies may be missed.

→ Too low: Too many false positives (normal points marked as anomalies).

- **What alternative methods can be used to define the threshold?**

→ Gaussian Mixture Models (GMM), Kernel Density Estimation (KDE), or domain-expert-defined thresholds.

5. Deployment & Real-time Monitoring

- **How is the trained model deployed?**

→ It is exported as an ONNX file and integrated into the Digital Twin for real-time inference.

- **How does the model handle real-time streaming data?**

→ Incoming heart rate data is preprocessed, fed into the LSTM, and compared to the learned patterns for anomaly detection.

- **What is the role of the Digital Twin in this system?**

→ It simulates a user's physiological state, monitoring real-time heart rate patterns and detecting abnormalities.

6. Model Performance & Improvement

- **What metrics can be used to evaluate this model?**
 - Reconstruction error, precision-recall, F1-score, and anomaly detection accuracy.
 - **How can the model be improved?**
 - By tuning hyperparameters, increasing dataset diversity, and experimenting with attention-based LSTMs.
 - **What are some limitations of this model?**
 - It assumes all normal patterns are well-represented in training, making it less effective for unseen heart rate trends.
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